Unit-2 Fuzzy Set and Fuzzy Logic

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(1) Fuzzy Sets & system :-

Def^n 2. The membership function μ_F of a fuzzy set F is a function

$$\mu_F : X \longrightarrow [0,1]$$

So, every element x from X has a membership degree $\mu_F(x) \in [0,1]$. F is completely determined by the set of tuples :

$$F = \{(x, \mu_F(x)) / x \in X\}$$

If $X = \{x_1, x_2, \cdots, x_n\}$, then a fuzzy set A of X could be written as

$$A = \{(x_1, \mu_A(x_1)), (x_2, \mu_A(x_2)), \cdots, (x_n, \mu_A(x_n))\}$$

Also written as;

$$A = \left\{ \frac{\mu_A(x_1)}{x_1}, \frac{\mu_A(x_2)}{x_2}, \cdots, \frac{\mu_A(x_n)}{x_n} \right\}$$

For eg. :- the real no. 3, 1 and 0 with membership grades 0.1, 0.09 and 1 can be written as;

$$A = \{(3, 0.1), (1, 0.09), (0, 1)\}$$

or $$A = \left\{ \frac{0.1}{3}, \frac{0.09}{1}, \frac{1}{0} \right\}$$

Crisp Set :- In fuzzy set theory, classical sets are called crisp sets.

A set defined using a characteristic function that assigns a value of either 0 or 1 to each element of the universe.

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thereby discriminating between members and non-members is called a crisp set

The membership function can be used to define a set A given by

$$\chi_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}$$

The function χ (read as chi) is known as membership function.

The set of natural numbers, real no., integers, rational numbers are all examples of crisp set.

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4.12. Fuzzy Sets

Computer is made up of circuits and devices. It has no intelligence of its own but it can be made intelligent artificially, it can be made expert in area like medical diagnosis, robot movement, location of mineral deposits etc. For this a mathematical movement of vague concepts or vague knowledge or imprecise data is necessary. In classical set theory, the "integer number which are greater than or equal to 5 and less than equal to 12" can be represented as

$$S = \{x \in Z : 5 \leq x \leq 12\}$$

i.e.,

$$S = \{5, 6, 7, 8, 9, 10, 11, 12\}$$

This representation is inadequate in the sense that we are not in a position to answer somewhat vogue concept *e.g.*, "integers which are more or less equal to 7" The source of vagueness is "more or less". In convention set theory the characteristic function

$$\mu : X -> \{0, 1\}$$

which associates with each element of a universe of discourse X either 1 or 0 which means that a particular element either belongs to the set or does not, respectively. Thus, there is a clear distinction of the elements 'belonging and not belonging to the set', or equivalently the transition from 'belonging to 'not belonging' to the set is abrupt. But for the "integers being more or less equal to 7", such a precise distinction is evidently artificial. Here it is not possible to fix any clear border line. One of the most important tool for such information processing is fuzzy set theory.

In fuzzy set theory, classical sets are called **crisp sets,** in order to distinguish them from fuzzy sets. Let C be a crisp set defined on the universe X, then for any element x of X, either $x \in C$ or $x \notin C$. In fuzzy set theory this property is generalised, therefore in a fuzzy set F, it is not necessary that either $x \in F$ or $x \notin F$. In fuzzy set theory, the characteristic function $\mu c : X \rightarrow \{0, 1\}$ defined in a crisp set is generalised to a membership function that assign to every $x \in X$, a value from the unit interval [0, 1] instead from the two-element set {0, 1}. The set that is defined on the basis of such an extended membership function is called a **fuzzy set.** A fuzzy set consists of elements and their respective grades of membership in the set. This grade corresponds to the degree to which that individual is similar or compatible with the concept represented by the fuzzy set. The grade of membership of an element is given by a subjectively defined membership function. The value of the grade of membership of an element can range from 0 to 1 where the value of 1 denotes full membership, and the closer the value is to 0, the weaker is the element's membership in the fuzzy set.

Definition: The **membership function** μ_F of a fuzzy set F is a function

$$\mu_F : X \rightarrow [0, 1].$$

So, every element x from X has a membership degree $\mu_F(x) \in [0, 1]$. F is completely determined by the set of tuples

$$F = \{(x, \mu_F(x) \mid x \in X\}.$$

Thus a fuzzy set is a set of pairs consisting of the particular element of the universe of discourse and their membership grades.

For example, one can define a possible membership function for the set of real numbers close to 0 as follows:

$$\mu_F(x) = \frac{1}{1+10x^2}$$

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The graph of the function is shown in Fig 4.3. We can determine the membership grade real numbers in this fuzzy set, which signifies the degree to which that number is close to instance, the number 3 is assigned a grade of 0.1, the number 1 a grade of .09, and the numb grade 1.

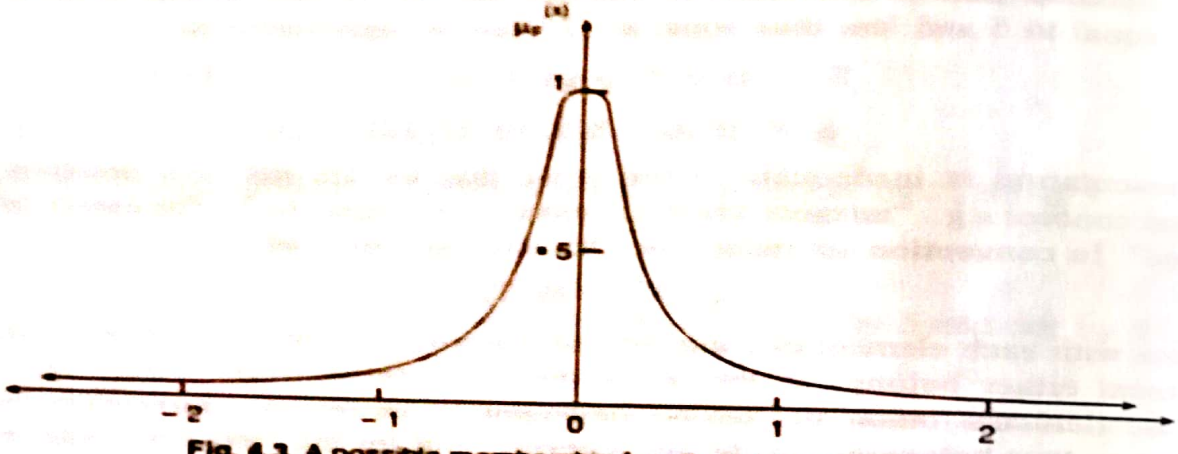


Fig. 4.3. A possible membership function of the fuzzy set of real numbers close to zero.

If $X = \{x_1, x_2, \ldots\ldots, x_n\}$, then a fuzzy set A of X could be written as

$$A = \{(x_1, \mu_A(x_1)), (x_2, \mu_A(x_2)) \ldots\ldots, (x_n, \mu_A(x_n))\}$$

which sometimes is written in the following way:

$$A = \left\{\frac{\mu_A(x_1)}{x_1}, \frac{\mu_A(x_2)}{x_2}, \ldots\ldots\ldots, \frac{\mu_A(x_n)}{x_n}\right\}$$

For example, the real numbers 3, 1 and 0 with membership grades 0.1, .09 and 1 can be written

$$A = \{(3, 0.1), (1, .09), (0, 1)\}$$

or

$$A = \left\{\frac{0.1}{3}, \frac{.09}{1}, \frac{1}{0}\right\}$$

The problem of constructing meaningful membership function in various contexts is a difficult task. A lot of research work is in progress in this area.

Some Useful Definitions

Containment

Let X be a set ($\neq \phi$) and A, B are two fuzzy sets of X with membership function μ_A and respectively. We say that the fuzzy set A is contained in the fuzzy set B if and only if

$$\mu_A(x) \leq \mu_B(x) \text{ for all } x \in X.$$

We may say, in another terminology, that A is a fuzzy subset of B, denoted by $A \subseteq B$

Example 44. If $X = \{1, 2, 3\}$ and A, B, C are three fuzzy sets given by

$A = \left\{\frac{.1}{1}, \frac{.5}{2}, \frac{1}{3}\right\}$ $B = \left\{\frac{.1}{1}, \frac{.4}{2}, \frac{.7}{3}\right\}$ and $C = \left\{\frac{.1}{1}, \frac{.6}{2}, \frac{.5}{3}\right\}$, then $B \subseteq A$, since $.1 = .1$, $.4 < .5$, $.7 < 1$ and $C \not\subseteq A$ since $.1 = .1$, $.6 > .5$ and $.5 < 1$

Normal fuzzy set

A fuzzy set A of a set X is called a normal fuzzy set if and only if

$$\max_{x \in X} \mu_A(x) = 1$$

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i.e., $\mu_A(x) = 1$ for the least one $x \in X$.

Otherwise A is **subnormal.**

Example 45. If $X = \{1, 2, 3, 4\}$ and $A = \left\{\frac{.1}{1}, \frac{1}{2}, \frac{.2}{3}, \frac{.5}{4}\right\}$ then A is a normal fuzzy set.

Support of a Fuzzy Set

The support of a fuzzy set A of the set X is the crisp set that contains all the elements of X that have a nonzero membership grade in A and denoted by supp (A) *i.e.*,

$$\text{supp (A)} = \{x \in X : \mu_A(x) > 0)\}$$

The element x in X at which $\mu_A(x) = 0.5$ is called the **crossover point.** A fuzzy set whose support is a single element in X with $\mu_A(x) = 1$ is called a **fuzzy singleton.** An empty fuzzy set has an empty support ; that is, the membership function assign 0 to all elements of the universal set.

Example 46. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and fuzzy set A of X is given by

$A = \{(1, 0), (2, 0), (3, 0.2), (4, 0.5), (5, 0.3), (6, 0.4), (7, 0) (8, 0), (9, 0)\}$

$= \{(3, 0.2), (4, 0.5), (5, 0.3), (6, 0.4)\}$ where $\mu_A(x) > 0$

Then supp (A) $= \{3, 4, 5, 6\}$

$x = 4$ is the crossover point.

A special notation is often used for defining fuzzy sets with a finite support. Assume that x_i is an element of the support of fuzzy set A and that μ_i is its grade of membership in A. Then A is written as $A = \mu_1 \mid x_1 + \mu_2 \mid x_2 + \ldots\ldots + \mu_n \mid x_n$.

where slash is used to link the elements of the support with their grades of membership in A and the plus sign indicates, rather than any sort of algebraic addition that the listed pairs of elements and memebrship grades collectively from the definition of the set A. For the case in which a fuzzy set A is defined on a universal set that is finite and countable, one may write

$$A = \Sigma\, \mu_i / x_i$$

α–cut or α–level set

The α-level set is the crisp set of elements that belong to the fuzzy set A at least to the degree α. In mathematical terms,

$$A_\alpha = \{x \in X : \mu_A(x) \geq \alpha\}.$$

The **strong α-level set** or **strong α-cut** is defined as

$$A'_\alpha = \{x \in X : \mu_A(x) > \alpha\}.$$

A fuzzy set is said to be **convex** if all α-level sets are convex.

Example 47. Suppose $X = \{1, 2, 3, 4, 5, 6\}$. Consider the fuzzy set A of X given by $A = \{(1, 0.2), (2, 0.5), (3, 0.7), (4, 1), (5, 0.8), (6, 0.3)\}$. Then all possible α-level sets are:

$A_{0.2} = \{1, 2, 3, 4, 5, 6\}$ $A_{0.3} = \{2, 3, 4, 5, 6\}$ $A_{0.5} = \{2, 3, 4, 5\}$

$A_{0.7} = \{3, 4, 5\}$ $A_{0.8} = \{4, 5\}$ $A_1 = \{4\}$.

Clearly, $A_0 = X$ and $A_\alpha = \phi$ for all $\alpha > 1$.

Basic operations of Fuzzy Sets

As in the conventional set theory, the basic operations in the theory of fuzzy sets are the complement, union and intersection. For brevity, the definitions are given in terms of the respective membership functions.

Equal Sets

Two fuzzy sets A and B are equal if

$$\mu_A(x) = \mu_B(x) \text{ for all } x \in X.$$

and is written as $A = B$. If for atleast one $x \in X$, $\mu_A(x) \neq \mu_B(x)$ then A and B are said to be unequal and written as $A \neq B$.

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For example, if $A = \{(x_1, 0.3), (x_2, 0.5)\}$, $B = \{(x_1, 0.2)\ (x_2, 0.5)\}$, $C = \{(x_1, 0.3), (x_2, 0.5)\}$

Then $A \neq B$ since $\mu_A(x_1) \neq \mu_B(x_1)$ though $\mu_A(x_2) = \mu_B(x_2)$

But $A = C$ since $\mu_A(x_1) = \mu_B(x_1) = 0.3$ and $\mu_A(x_2) = \mu_B(x_2) = 0.5$

Complement

The complement of a fuzzy set A is denoted by A^c (or A') and is defined by its membership function as $\mu_{A^c}(x) = 1 - \mu_A(x)$ for all x.

Thus if an element has a membership grade of 0.7 in a fuzzy set A, its membership grade in the complement of A will be 0.3.

Example 48. If $A = \{(x_1, 0), (x_2, 0.3), (x_3, 0.5)\}$

Then $A^c = \{(x_1, 1), (x_2, 0.7), (x_3, 0.5)\}$

since $\mu_A{}^c(x_1) = 1 - \mu_A(x_1) = 1 - 0 = 1$; $\mu_A{}^c(x_2) = 1 - \mu_A(x_2) = 1 - 0.3 = 0.7$

and $\mu_A{}^c(x_3) = 1 - \mu_A(x_3) = 1 - 0.5 = 0.5$

Union

The union of two fuzzy sets A and B is a fuzzy set C given by

$$C = A \cup B$$

where $\mu_c(x) = \max [\mu_A(x), \mu_B(x)]; x \in X$

Example 49. If $A = \{(4, 01), (6, 05), (8, 0.6), (10, 0.7)\}$

$B = \{(4, 02), (6, 1), (8, 04), (10, 0.5)\}$

Then $C = A \cup B = \{(4, 02), (6, 1), (8, 0.6), (10, 0.7)\}$

since $\mu_C(x_1) = \max [\mu_A(x_1), \mu_B(x_1)] = \max [0.1, 0.2] = 0.2$;

$\mu_C(x_2) = \max [\mu_A(x_2), \mu_B(x_2)] = \max [0.5, 1] = 1$;

$\mu_C(x_3) = \max [\mu_A(x_3), \mu_B(x_3)] = \max [0.4, 0.6] = 0.6$;

$\mu_C(x_4) = \max [\mu_A(x_4), \mu_B(x_4)] = \max [0.7, 0.5] = 0.7$.

Example 50. If $X = \{51, 52, 53, 54\}$

$$A = \frac{0.2}{51} + \frac{0.5}{52} + \frac{0.8}{53} + \frac{1}{54}$$

and

$$B = \frac{1}{51} + \frac{0.8}{52} + \frac{0.5}{53} + \frac{0.2}{54}$$

then what is the value of $A \cup B$?

Solution. $A \cup B = \max [0.2, 1]/51 + \max [0.5, 0.8]/52 + \max (0.8, 0.5]/53 + \max [1, 02]/54$

$= 1/51 + 0.8/52 + 0.8/53 + 1/54$

$= \{(51, 1), (52, .8), (53, .8), (54, 1)\}$

Intersection

The intersection of two fuzzy sets A and B is a fuzzy set C given by

$$C = A \cap B$$

where $\mu_C(x) = \min [\mu_A(x), \mu_B(x)]; x \in X$

Example 51. If $A = \{(3, 0.1), (5, 0.7), (7, 0.7)\}$

$B = \{(3, 0.4), (5, 0.8), (7, 0.3)\}$

Then $C = A \cap B = \{(3, 0.1), (5, 0.7), (7, 03)\}$

since $\mu_C(x_1) = \min [\mu_A(x_1), \mu_B(x_1)] = \min [0.1, 0.4] = 0.1$;

$\mu_C(x_2) = \min [\mu_A(x_2), \mu_B(x_2)] = \min [0.7, 0.8] = 0.7$;

$\mu_C(x_3) = \min [\mu_A(x_3), \mu_B(x_3)] = \min [0.7, 0.3] = 0.3$.

These formulation of fuzzy complement, union and intersection perform identically to the corresponding crisp set operations when membership grades are restricted to the values 0 and 1.

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They are therefore, generation of the classical crisp set operations. Sometimes it is more convenient to give the graphs that represent the membership functions as in Fig. 4.4.

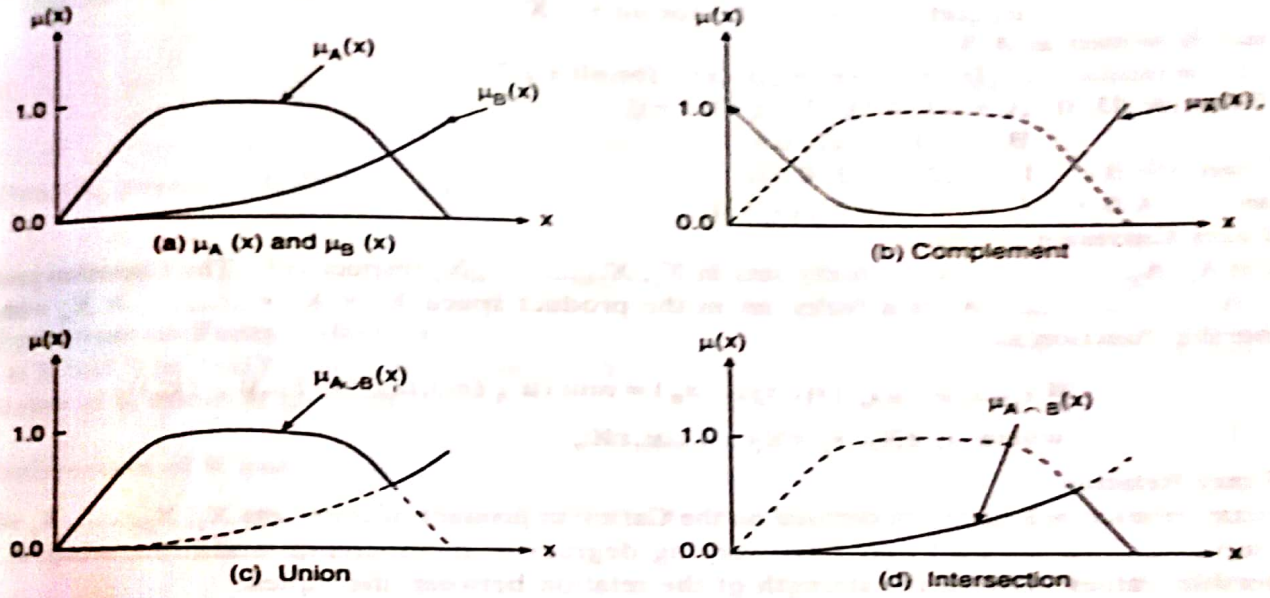


Fig. 4.4

Difference

The difference of two fuzzy sets A and B is defined by

$$A - B = A \cap B^c$$

Example 52. If $A = \{(x_1, 0.3), (x_2, 0.4), (x_3, 0.5)\}$

$B = \{(x_1, 0.2), (x_2, 0.6), (x_3, 0.7)\}$

Then $B^c = \{x_1, 0.8), (x_2, 0.4), (x_3, 0.3)\}$

So $A - B = A \cap B^c = \{x_1, 0.3), (x_2, 0.4), (x_3, 0.3)\}$

Note that, except in particular cases $A - B \neq B - A$

Properties of Fuzzy set operations

The properties of fuzzy set operations that are common to crisp set operations are

Idempotent : $A \cup A = A, A \cup A = A$

Commutative : $A \cup B = B \cup A, A \cap B = B \cap A$

Associative : $(A \cup B) \cup C = A \cup (B \cup C), (A \cap B) \cap C = A \cap (B \cap C)$

Distributive : $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Absorption : $A \cup (A \cap B) = A, A \cap (A \cup B) = A$

Identity : $A \cup \varphi = A, A \cap U = A$

Double negation law (Involution)

If A is a fuzzy set, then $(A^c)^c = A$

De Morgan's law

If A and B are two fuzzy sets then

$$(A \cup B)^c = A^c \cap B^c; (A \cap B)^c = A^c \cup B^c$$

Algebraic Sum

The algebraic sum of two fuzzy sets A and B is defined by the membership function as

$$\mu_{A.B}(x) = \mu_A(x) + \mu_B(x)\ \mu_A(x)\ \mu_B(x) \text{ for all } x \varepsilon X$$

and written as A.B.

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Algebraic Product

The algebraic product of two fuzzy sets A and B is defined by two membership functions as

$$\mu_{A.B}(x) = \mu_A(x) \cdot \mu_B(x) \text{ for all } x \varepsilon X$$

and is written as A.B

In particular $\mu_{A.A}(x) = \mu_A{}^2(x) = [\mu_A(x)]^2$ for all $x \varepsilon X$

Example 53. If A = {(1, 0.5), (2, 1), (3, 0.6)}

B = {(1, 1), (2, 0.6)}

Then A + B = {(1, 1), (2, 1), (3, 0.6)}

and A.B = {(1, 0.5), (2, 0.6), (3, 0)}

Fuzzy Cartesian product

Let $A_1, A_2, \ldots\ldots\ldots\ldots, A_n$ be fuzzy sets in $X_1, X_2, \ldots\ldots\ldots, X_n$ respectively. The Cartesian product of $A_1, A_2, \ldots\ldots\ldots\ldots, A_n$ is a fuzzy set in the product space $X_1 \times X_2 \times\ldots\ldots\ldots\ldots \times X_n$ with the membership function as.

$$\mu_{A_1 \times A_2 \times \ldots \times A_n}(x_1, x_2, \ldots x_n) = \min(\mu_{A_1}(x_1), \mu_{A_2}(x_2) \ldots, \mu_{A_n}(x_n))$$

where $x_1 \varepsilon X_1, x_2 \varepsilon X_2, \ldots\ldots x_n \varepsilon X_n$

Fuzzy Relation

Fuzzy relation is a fuzzy set defined on the Cartesian product of crisp sets $X_1, X_2, \ldots\ldots, X_n$ where the n-tuples $(x_1, x_2, \ldots\ldots, x_n)$ may have varying degrees of membership within the relation. The membership values indicate the strength of the relation between the tuples.

Binary Relations

Any relation between two sets X and Y is known as binary relation. The domain of a binary fuzzy relation R(X, Y) is the fuzzy set dom R(X, Y) with the membership function

$$\mu_{\text{dom } R}(x) = \max_{y \varepsilon Y} \mu_R(x, y) \text{ for each } x \varepsilon X$$

The range of a binary fuzzy relation R (X, Y) is the fuzzy set ran R (X, Y) with the membership function

$$\mu_{\text{ran } R}(y) = \max_{x \varepsilon X} \mu_R(x, y) \text{ for each } y \varepsilon Y$$

The height of a fuzzy relation R is a number h (R) defined by

$$h(R) = \max_{y \in Y} \max_{x \in X} \mu_R(x, y)$$

that is, the largest membership grade attained by any pair (x, y) in R. If $h(R) = 1$, the fuzzy relation R is called normal, otherwise it is called subnormal.

The inverse of a fuzzy relation R is defined by

$$\mu_{R^{-1}}(y, x) = \mu_R(x, y) \text{ for all } (x, y) \in X \times Y$$

A fuzzy binary relation can be represented as a two dimensional membership array whose entries correspond to ordered pairs in the universal set. These entries take values representing the membership grades of the corresponding ordered pairs.

Let $X = \{x_1, x_2, \ldots\ldots\ldots\ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots\ldots\ldots\ldots, y_m\}$

The fuzzy relation can be expressed by an $n \times m$ matrix (known as membership matrix) as

$$M_R = R(X, Y) = \begin{bmatrix} \mu_R(x_1, y_1) & \mu_R(x_1, y_2) \ldots\ldots\ldots\ldots \mu_R(x_1, y_m) \\ \mu_R(x_2, y_1) & \mu_R(x_2, y_2) \ldots\ldots\ldots\ldots \mu_R(x_2, y_m) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \mu_R(x_n, y_1) & \mu_R(x_n, y_2) \ldots\ldots\ldots\ldots \mu_R(x_n, y_m) \end{bmatrix}$$

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Example 54. Let R(X, Y) be a fuzzy relation on $X = \{x_1, x_2, x_3\}$ and $Y = \{y_1, y_2\}$ and a fuzzy binary relation is given by membership matrix as

$$M_R = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 1 & .2 \\ .5 & 0 \\ .4 & .3 \end{bmatrix} \end{array}$$

Then the inverse of R (X, Y) is given by

$$M_{R^{-1}} = M_R^T = \begin{array}{c} \\ y_1 \\ y_2 \end{array} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 1 & .5 & .4 \\ .2 & 0 & .3 \end{bmatrix} \end{array}$$

Operations on Fuzzy Relations

Let R and S be two fuzzy relations on X × Y.

Union of R and S is given by

$$\mu_{R \cup S}(x, y) = \max(\mu_R(x, y), \mu_S(x, y))$$

Intersection of R and S is given by

$$\mu_{R \cap S}(x, y) = \min(\mu_R(x, y), \mu_s(x, y))$$

Complement of R is given by

$$\mu_{R^C}(x, y) = 1 - \mu_R(x, y)$$

Algebraic products is given by

$$\mu_{R.S}(x, y) = \mu_R(x, y).\ \mu_S(x, y)$$

Algebraic sum is given by

$$\mu_{R+S}(x, y) = \mu_R(x, y) + \mu_S(x, y) - \mu_R(x, y).\ \mu_S(x, y)$$

Composition of Relations

Suppose R is a fuzzy relation defined on X × Y, and S is a fuzzy relation defined on Y × Z, then R o S is a fuzzy relation on X × Z. The fuzzy max-min composition is defined as

$$\mu_{R \circ S}(x, z) = \max_{y \in Y} (\min(\mu_R(x, y), \mu_S(y, z))$$

Example 55. Let R and S be two fuzzy relations from X to Y given in the following matrix forms. Find

(*a*) R ∪ S (*b*) R ∩ S (*c*) R + S (*d*) R.S (*e*) R^C.

$$M_R = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .3 & 1 & .2 \\ .8 & 0 & .5 \end{bmatrix} \end{array}; \quad M_R = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .6 & .1 & .9 \\ 0 & .2 & .3 \end{bmatrix} \end{array}$$

Solution:

(*a*) $$M_{R \cup S} = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .6 & 1 & .9 \\ .8 & 2 & .5 \end{bmatrix} \end{array}$$

(*b*) $$M_{R \cap S} = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .3 & .1 & .2 \\ .0 & 0 & .3 \end{bmatrix} \end{array}$$

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$$(c)\quad M_{R+S} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .72 & .1 & .92 \\ .8 & .2 & .65 \end{bmatrix} \end{array}$$

$$(d)\quad M_{R\,S} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .18 & .1 & .18 \\ 0 & 0 & .15 \end{bmatrix} \end{array}$$

$$(e)\quad M_{R^C} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .7 & 0 & .8 \\ .2 & 0 & .5 \end{bmatrix} \end{array}$$

Example 56. Let $X = \{x_1, x_2\}$. $Y = \{y_1, y_2, y_3\}$ and $Z = \{z_1, z_2, z_3, z_4\}$. Let R and S be fuzzy relation on X×Y and Y×Z respectively, given below in the matrix form as

$$M_R = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} .3 & .8 & 1 \\ .9 & .7 & .4 \end{bmatrix} \end{array};\quad M_s = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array}\begin{array}{c} \begin{array}{cccc} z_1 & z_2 & z_3 & z_4 \end{array} \\ \begin{bmatrix} .7 & .6 & .4 & .1 \\ .4 & .1 & .7 & .2 \\ .5 & .9 & .6 & .8 \end{bmatrix} \end{array}$$

Find the composition of R and S in the matrix form.

Solution:

$$\mu_{Ros}(x_1, z_1) = \max(\min(\mu_R(x_1, y_1), \mu_s(y_1, z_1)), \min(\mu_R(x_1, y_2), \mu_s(y_2, z_1)), \min(\mu_R(x_1, y_3), \mu_s(y_3, z_1)))$$
$$= \max(\min(.3, .7), \min(.8, .4), \min(1, .5))$$
$$= \max(.3, .4, .5) = .5$$

$$\mu_{Ros}(x_1, z_2) = \max(\min(\mu_R(x_1, y_1), \mu_s(y_1, z_2)), \min(\mu_R(x_1, y_2), \mu_s(y_2, z_2)), \min(\mu_R(x_1, y_3), (\mu_s(y_3, z_2)))$$
$$= \max(\min(.3, .6), \min(.8, .1), \min(1, .9))$$
$$= \max(.3, .1, .9) = .9$$

$$\mu_{Ros}(x_1, z_3) = \max(\min(\mu_R(x_1, y_1), \mu_s(y_1, z_3)), \min(\mu_R(x_1, z_2), \mu_z(y_2, z_3)), \min(\mu_R(x_1, y_3), \mu_s(y_3, z_3)))$$
$$= \max(\min(.3, .4), \min(.8, .7), \min(1, .6))$$
$$= \max(.3, .7, .6) = .7$$

$$\mu_{Ros}(x_1, z_4) = \max(\min(\mu_R(x_1, y_1), \mu_s(y_1, z_4)), \min(\mu_R(x_1, y_2), \mu_s(y_2, z_4)), \min(\mu_R(x_1, y_3), \mu_s(y_3, z_4)))$$
$$= \max(\min(.3, .1), \min(.8, .2), \min(1, .8))$$
$$= \max(.1, .2, .8) = .8$$

Similarly,

$$\mu_{Ros}(x_2, z_1) = .7,\ \mu_{Ros}(x_2, z_2) = .6$$
$$\mu_{Ros}(x_2, z_3) = .7,\ \mu_{Ros}(x_2, z_4) = .4$$

$$M_{Ros} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{cccc} z_1 & z_2 & z_3 & z_4 \end{array} \\ \begin{bmatrix} .5 & .9 & .7 & .8 \\ .7 & .6 & .7 & .4 \end{bmatrix} \end{array}$$

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Fuzzy Relations

In this section we shall discuss the concept of Fuzzy Relation. Before we discuss this concept we first recapitulate crisp relation over crisp sets.

Let X and Y be any two ordinary (crisp) sets. Recall that the cartesian product of X and Y, denoted by $X \times Y$, is the set of all ordered pairs of elements of X and Y. More precisely

$$X \times Y = \{ (x,y) : x \in X, y \in Y\}.$$

Observe that if $X \neq Y$, then $X \times Y \neq Y \times X$. The cartesian product can be generalised to number of sets. Let $X_1, X_2, \ldots, X_n$ be n sets, then

$$X_1 \times X_2 \times \ldots \times X_n = \underset{i=1,2,\ldots,n}{X} X_i$$
$$= \{(x_1, x_2, \ldots, x_n); x_i \in X_i, i = 1, 2, \ldots, n\}.$$

A crisp binary relation R is a subset of $X \times Y$. Thus

$$R = R(X \times Y) \subseteq X \times Y.$$

A binary relation R can be expressed as a characteristic function in the following way

$$\mu_R (x,y) = \begin{cases} 1 \text{ if } (x, y) \in R \\ 0 \text{ otherwise.} \end{cases}$$

Similarly an n-ary relation R is a subset of $X_1 \times X_2 \times \ldots \times X_n$ and such a relation can also be expressed as a characteristic function as follows:

$$\mu_R (x_1, x_2, \ldots, x_n) = \begin{cases} 1 \text{ if } (x_1, x_2, \ldots, x_n) \in R \\ 0 \text{ otherwise.} \end{cases}$$

Fuzzy Relations

We now introduce the concept of fuzzy relation. Let X and Y be two sets. A fuzzy (binary) relation R is a fuzzy subset of the crisp cartesian product set $X \times Y$. Thus the ordered pair (x,y) may have varying degrees of membership taking values in the closed interval $[0,1]$. Similarly an n-ary fuzzy

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relation R is a fuzzy subset of $X_1 \times X_2 \times \ldots \times X_n$ where the n-tuple $(x_1, x_2, \ldots, x_n)$ may have varying degrees of membership and the membership grade is usually represented by a real number in the closed interval [0, 1].

A fuzzy binary relation can be conveniently represented as a two-dimensional membership array whose entries correspond to ordered pairs in the universal set. These entries take values representing the membership grades of the corresponding ordered pairs. Similarly a fuzzy n-ary relation can be expressed as n-dimensional membeship array.

An example of a fuzzy relation $\underline{R}$ can be given in a two-dimensional array in the following manner.

	y_1	y_2	y_3	y_4	y_5
x_1	0	0	0.1	0.3	1
x_2	0	0.8	0	0	1
x_3	0.4	0.4	0.5	0	0.2

From the above table it means, for example, that $(x_1, y_3) \in \underline{R}$ with the degree or grade of membership $\mu_{\underline{R}}(x_1, y_3) = 0.1$.

We now recall the definitions of domain, range of a crisp binary relation R and the inverse of R.

Let $R = R(X,Y)$ be a crisp binary relation. Then

$$\text{dom } R(X, Y) = \{x \in X : (x, y) \in R \text{ for some } y \in Y\}.$$

$$\text{rng } R(X, Y) = \{y \in Y : (x, y) \in R \text{ for some } x \in X\}.$$

$$R^{-1}(X, Y) = \{(y, x) : (x, y) \in R\}.$$

With these motivations the domain, range and inverse of fuzzy relations are defined.

Let $\underline{R}(X,Y)$ be a fuzzy binary relation. Its domain is the fuzzy set dom $\underline{R}(X,Y)$ whose membership function is given by

$$\mu_{\text{dom } \underline{R}}(x) = \max_{y \in Y} \mu_{\underline{R}}(x, y),$$

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for each $x \in X$. The range is the fuzzy set rng $\underline{R}(X,Y)$ whose membership function is given by

$$\mu_{\text{rng}\,\underline{R}}(y) = \max_{x \in X} \mu_{\underline{R}}(x,y)$$

for each $y \in Y$. We now define the concept of height of a fuzzy relation $\underline{R}$. The height of $\underline{R}$ is a number $h(\underline{R})$ given by

$$h(\underline{R}) = \max_{y \in Y} \max_{x \in X} \mu_{\underline{R}}(x, y).$$

It is the largest membership grade attained by any pair (x,y) in $\underline{R}$. If $h(\underline{R}) = 1$, the fuzzy relation $\underline{R}$ is called normal, otherwise it is called subnormal.

The inverse of a fuzzy relation $\underline{R}$ is given by

$$\mu_{\underline{R}^{-1}}(y, x) = \mu_{\underline{R}}(x, y).$$

If a binary fuzzy relation is given by

	y_1	y_2
x_1	.3	.2
x_2	0	1
x_3	.6	.4

Then the inverse is given by

	x_1	x_2	x_3
y_1	.3	0	.6
y_2	.2	1	.4

We shall now discuss some concepts related to fuzzy binary relations $\underline{R}$.

Support of a fuzzy relation. The support $s(\underline{R})$ of a fuzzy binary relation $\underline{R}$ is the ordinary subset of ordered pairs for which the membership function is non-zero. More precisely,

$$s(\underline{R}) = \{(x, y) : \mu_{\underline{R}}(x, y) > 0\}.$$

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An example, let the following be a fuzzy relation.

	y_1	y_2	y_3	y_4
x_1	.1	0	.2	0
x_2	.3	0	0	.9
x_3	.4	.7	1	1

Then $s(\underline{R})$ is given by

$$s(\underline{R}) = \{(x_1,y_1), (x_1,y_3), (x_2, y_1), (x_2, y_4), (x_3, y_1), (x_3, y_2), (x_3, y_3), (x_3, y_4)\}.$$

Ordinary relation closest to a fuzzy relation. Let $\underline{R}$ be a fuzzy relation. An ordinary (crisp) relation $\underline{\underline{R}}$ closest to $\underline{R}$ is given by

$$\mu_{\underline{\underline{R}}}(x,y) = \begin{cases} 0 \text{ if } \mu_{\underline{R}}(x, y) < .5, \\ 1 \text{ if } \mu_{\underline{R}}(x, y) > .5, \\ 0 \text{ or } 1 \text{ if } \mu_{\underline{R}}(x, y) = .5. \end{cases}$$

Usually we take $\mu_{\underline{\underline{R}}}(x,y) = 0$ if $\mu_{\underline{R}}(x,y) = .5$.

The following example shows how we pass from $\underline{R}$ to $\underline{\underline{R}}$.

$\underline{R}$

	y_1	y_2	y_3	y_4	y_5	y_6
x_1	.7	.3	.2	1	0	.8
x_2	.5	.4	0	.6	.9	.1
x_3	.6	1	.8	0	0	.7

$\underline{\underline{R}}$

	y_1	y_2	y_3	y_4	y_5	y_6
x_1	1	0	0	1	0	1
x_2	0	0	0	1	1	0
x_3	1	1	1	0	0	1

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Ordinary relation (subset) of level α in a fuzzy relation. Let $\alpha \in [0, 1]$ and $\underline{R}$ a fuzzy binary relation from X to Y. The ordinary subset of level α of $\underline{R}$ is given by the ordinary subset

$$G_\alpha = \{(x,y) : \mu_{\underline{R}}(x,y) \geq \alpha\}.$$

Let $\alpha = .8$ and $\underline{R}$ be given by

	y_1	y_2	y_3	y_4
x_1	.3	.8	1	0
x_2	.5	1	.3	.9
x_3	1	.2	.6	.7

We have

$$G_\alpha = G_{.8} = \{(x_1,y_2), (x_1, y_3), (x_2, y_2), (x_2, y_4), (x_3, y_1)\}.$$

Algebra of Fuzzy Relations

Since fuzzy relations are fuzzy subsets of the cartesian product set, we can talk of union, intersection, algebraic sum, algebraic product and composition of fuzzy relations and the complement of a fuzzy relation.

Let $\underline{R}$ and $\underline{S}$ be two fuzzy (binary) relations from X to Y.

Union of $\underline{R}$ and $\underline{S}$ is given by

$$\mu_{\underline{R} \cup \underline{S}}(x, y) = \max \{\mu_{\underline{R}}(x,y), \mu_{\underline{S}}(x,y)\}.$$

Intersection is given by

$$\mu_{\underline{R} \cap \underline{S}}(x, y) = \min \{\mu_{\underline{R}}(x, y), \mu_{\underline{S}}(x, y)\}.$$

Algebraic product is

$$\mu_{\underline{R} \cdot \underline{S}}(x, y) = \mu_{\underline{R}}(x, y)\, \mu_{\underline{S}}(x, y)$$

and algebraic sum is given by

$$\mu_{\underline{R} + \underline{S}}(x, y) = \mu_{\underline{R}}(x, y) + \mu_{\underline{S}}(x, y) - \mu_{\underline{R}}(x, y)\, \mu_{\underline{S}}(x, y).$$

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The complement of $\underline{R}$ is given by

$$\mu_{\underline{R}^c}(x, y) = 1 - \mu_{\underline{R}}(x, y).$$

Let $\underline{R}$ and $\underline{S}$ be two fuzzy binary relations given in the following two-dimensional arrays.

$\underline{R}$

	y_1	y_2	y_3	y_4
x_1	.3	.2	1	0
x_2	.8	1	0	.2
x_3	.5	0	.4	0

$\underline{S}$

	y_1	y_2	y_3	y_4
x_1	.3	0	.7	0
x_2	.1	.8	1	1
x_3	.6	.9	.3	.2

Then $\underline{R} \cup \underline{S}$ is

	y_1	y_2	y_3	y_4
x_1	.3	.2	1	0
x_2	.8	1	1	1
x_3	.6	.9	.4	.2

$\underline{R} \cap \underline{S}$ is given by

	y_1	y_2	y_3	y_4
x_1	.3	0	.7	0
x_2	.1	.8	0	.2
x_3	.5	0	.3	0

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$\underline{R} \cdot \underline{S}$ is given by

	y_1	y_2	y_3	y_4
x_1	.09	0	.7	0
x_2	.08	.8	0	.2
x_3	.3	0	.12	0

and algebraic sum $\underline{R} + \underline{S}$ is given by

	y_1	y_2	y_3	y_4
x_1	.51	.20	1	0
x_2	.82	1	1	1
x_3	.80	.90	.58	.20

The complement of $\underline{R}$ is given by

	y_1	y_2	y_3	y_4
x_1	.7	.8	0	1
x_2	.2	0	1	.8
x_3	.5	1	.6	1

Fuzzy Relation Defined on a Set

We shall now discuss fuzzy binary relation defined on a single set (from a set to itself) and reflexivity, symmetry, antisymmetry and transitivity of such relations. We shall first recall these concepts for crisp relations defined on a set.

Let X be a set and R a crisp relation defined on it. R is called reflexive iff $(x, x) \in R$ for all $x \in X$, symmetry iff $(x, y) \in R \Rightarrow (y, x) \in R$, antisymmetric iff $(x, y) \in R$, $(y, x) \in R \Rightarrow x = y$, transitive iff $(x, y) \in R$, $(y, z) \in R \Rightarrow (x, z) \in R$.

These concepts can be extended to fuzzy binary relation defined on a single set by defining them in terms of membership functions.

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Let $\underline{R}$ be a fuzzy relation defined on a set X. R is called

reflexive iff $\mu_{\underline{R}}(x, x) = 1$ for all $x \in X$,

symmetric iff $\mu_{\underline{R}}(x, y) = \mu_{\underline{R}}(y, x)$,

antisymmetric iff $\mu_{\underline{R}}(x, y) > 0, \mu_{\underline{R}}(y, x) > 0\ 0 \Rightarrow x = y$,

transitive iff

$$\mu_{\underline{R}}(x, z) \geq \max_{y \in Y} \min[\mu_{\underline{R}}(x, y), \mu_{\underline{R}}(y, z)].$$

Transitivity is also called max-min transitivity.

Examples

1. Let $\underline{R}$ be the fuzzy relation 'very near' defined on the set of cities.
   A city is certainly very near to itself with degree of membership 1, the relation is reflexive. Further if a city A is very near to B with some degree, B is certainly very near to A with the same degree and hence the relation is symmetric. But observe that this relation is not transitive.

2. The following fuzzy relation is reflexive.

	x_1	x_2	x_3	x_4
x_1	1	0	0.2	0.3
x_2	0	1	0.1	1
x_3	0.2	0.7	1	0.4
x_4	0	1	0.4	1

3. The following one is symmetric

	x_1	x_2	x_3	x_4	x_5
x_1	0	.1	0	0.1	0.9
x_2	.1	1	.2	0.3	0.4
x_3	0	.2	.8	0.8	1
x_4	.1	.3	.8	0.7	1
x_5	.9	.4	1	1	0

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4. The fuzzy relation 'x resembles y' is reflexive, symmetric but not transitive.
5. The following are some examples of fuzzy transitive relations.

y is much larger than x,
x is cleaner than y,
x is a distant relative of y.

We define various types of crisp relations such as : equivalence, quasi-equivalence, compatibility or tolerance, partial ordering and pre-order or quasi-order; of which two namely equivalence and partial order are most important and useful. In a similar vein we define all such concepts for fuzzy relations. What is known as equivalence relation in case of crisp ones, the corresponding counterpart in the case of fuzzy relation is called the similarity relation or the fuzzy relation of similitude. We now express all these concepts in a tabular form.

	Reflexive	*Symmetric*	*Antisymmetric*	*Transitive*
Crisp: equivalence	√	√		√
Fuzzy: Similarity	√	√		√
Quasi-equivalence		√		√
Compatibility or Tolerance	√	√		
Partial order	√		√	√
Pre order or quasi-order	√			√

Similarity Relation

We shall now discuss the concept of fuzzy equivalence relation. Fuzzy equivalence relations are usually called similarity relations or fuzzy relations of similitude. Before we discuss this we first recall the concept of (crisp) equivalence relation.

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*A crisp binary relation is called an equivalence relation if it is reflexive, symmetric and transitive. Recall that a beautiful property, of such a relation is that it defines a partition on the whole set X. For $x \in X$ define a set A_x by

$$A_x = \{y : (x, y) \in R\}.$$

Since R is reflexive A_x is a non-empty subset of X. Because of symmetry and transitivity, each member of A_x is related to every other member of A_x and no member of A_x is related to any member not belonging to A_x. Each such set A_x, for each $x \in X$, is called an equivalence class or a partitioned class or a block. Two such classes are either identical or disjoint and union of all such classes is X. The family of all such classes is usually denoted by X/R and this forms a partition of X.

With these ideas we now discuss the concept of similarity relation; it is reflexive, symmetric and transitive. We usually use max-min form of transitivity.

The following are examples of similarity relations.

	a	b	c	d	e	f	g
a	1	.8	0	.4	0	0	0
b	.8	1	0	.4	0	0	0
c	0	0	1	0	1	.9	.5
d	.4	.4	0	1	0	0	0
e	0	0	1	0	1	.9	.5
f	0	0	.9	0	.9	1	.5
g	0	0	.5	0	.5	.5	1

	a	b	c	d	e
a	1	.8	.7	1	.9
b	.9	1	.7	.8	.8
c	.7	.7	1	.7	.8
d	1	.8	.7	1	.9
e	.9	.8	.7	.9	1

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We have observed that an equivalence relation clearly partitions elements into disjoint classes. But this is not true for a similarity relation. The case of a similarity relation can be approached in two different ways. Firstly, the similarity relation can be considered to partition elements into crisp sets whose members are 'similar' to each other to some specified degree. When this degree is 1, the partitioning is an equivalence class. Secondly, we may consider the degree of similarity that the elements of X have to some specific element $x \in X$. Thus, for each $x \in X$ a similarity class can be defined as a fuzzy set in which the membership grade of any particular element represents the similarity of that element to the element x. If all the elements in the class are similar to x of degree 1 and similar to all elements outside the set to the degree 0, then the partitioning is an equivalence class. For a similarity relation similarity class for each $x \in X$ can be defined as a fuzzy set in which the membership grade of each element $y \in X$ is the strength of that element's relation to x similarity classes are not in general disjoint.

We now discuss fuzzy compatibility relation. A fuzzy binary relation is called a compatibility or tolerance relation if it is reflexive and symmetric. This is also some times called a proximity relation.

The following fuzzy relation is a compatibility relation.

	a	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*
a	1	.8	0	0	0	0	0	0	0
b	.8	1	0	0	0	0	0	0	0
c	0	0	1	1	.8	0	0	0	0
d	0	0	1	1	.8	.7	.5	0	0
e	0	0	.8	.8	1	.7	.5	.7	0
f	0	0	0	.7	.7	1	.4	0	0
g	0	0	0	.5	.5	.4	1	0	0
h	0	0	0	0	.7	0	0	1	0
i	0	0	0	0	0	0	0	0	1

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Fuzzy Order Relation

We shall now discuss the other important and useful fuzzy binary relation namely fuzzy preorder and order relations. A binary fuzzy relation is called fuzzy preorder if it is reflexive and transitive. The following figure represents a fuzzy preorder on the set, $\{a, b, c, d, e\}$.

	a	b	c	d	e
a	1	.7	.8	.5	.5
b	0	1	.3	0	.2
c	0	.7	1	0	.2
d	.6	1	.9	1	.6
e	0	0	0	0	1

A binary fuzzy relation is called a fuzzy order relation if it is reflexive, antisymmetric and transitive. This is also called a fuzzy partial order relation. In other words, a fuzzy preorder relation which is antisymmetric is a fuzzy partial order relation.

The following is an example of a fuzzy partial order relation.

	a	b	c	d	e
a	1	.7	0	1	.7
b	0	1	0	.9	0
c	.5	.7	1	1	.8
d	0	0	0	1	0
e	0	.1	0	.9	1

Any fuzzy partial order relation can be resolved into a series of crisp partial orderings in the same manner in which this is done for the similarity relations. *A* set X along with a fuzzy partial ordering defined on it is called a fuzzy partially ordered set (fuzzy poset in short). Two fuzzy sets are associated with each element $x \in X$ in a fuzzy poset. One of them is called the dominating class of x. It is denoted by $R \geq[x]$ and is defined by

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$$\mu R \geq [x]^{(y)} = \mu R(x,y)$$

where $y \in X$. In otherwords, the dominating class of x contains the members of X to the degree to which they dominate x.

The other fuzzy set is the class dominated by x, which is denoted by $\mu R_{\leq}[x]$ and is defined by

$$\mu R\ [x] \leq {}^{(y)} = \mu R(y, x)$$

where $y \in X$. The class dominated by x contains the elements of X to the degree to which they are dominated by x. An element $x \in X$ is said to be undominated iff

$$\mu R(x, y) = 0$$

for all $y \in X$ and $x \neq y$. $x \in X$, is said to be undominating iff

$$\mu R(y, x) = 0$$

for all $y \in X$ and $y \neq x$.

Let X be a fuzzy poset with a fuzzy partial order R and A a crisp subset of X. The fuzzy upper bound for A is the fuzzy set denoted by $U(R, A)$ and is defined by

$$U(R, A) = \bigcap_{x \in A} {}^{R}\geq [x]$$

where $\cap$ denotes fuzzy intersection.

If a least upper bound of the set A exists, it is the unique element x in $U(R, A)$ such that

$$\mu U(R,A)^{(x)} > 0 \text{ and } \mu R(x,y) > 0$$

for all y in the support of $U(R, A)$. In the similar manner we define fuzzy lower bound and the greatest lower bound of the set A. These definitions reduce to those of usual upper bound and lower bound when the partial ordering is crisp.

As in the crisp case we can also define fuzzy linear order and fuzzy lattice.

A fuzzy linear order R on a set X is a fuzzy partial order such that for all x, y if $x \neq y$, then either $R(x, y) > 0$ or $R(y, x) > 0$.

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A fuzzy poset X is called a fuzzy lattice if every two element nonfuzzy set in X has fuzzy lower bound and fuzzy upper bound.

Exercises

1. Find $\underset{\sim}{A} \cup \underset{\sim}{B}$, $\underset{\sim}{A} \cap \underset{\sim}{B}$, $\underset{\sim}{A} - \underset{\sim}{B}$ and $\underset{\sim}{B} - \underset{\sim}{A}$ in the following cases:
   i) $\underset{\sim}{A} = \{(x_1 \mid 0.2), (x_2 \mid 0.8), (x_3 \mid 0.6)\}$
   $\underset{\sim}{B} = \{(x_1 \mid 0.4), (x_2 \mid 0.6), (x_3 \mid 1.0)\}$.
   ii) $\underset{\sim}{A} = \{(x_1 \mid 1.0), (x_2 \mid 0.0), (x_3 \mid 0.5), (x_4 \mid 0.7)\}$
   $\underset{\sim}{B} = \{(x_1 \mid 0.8), (x_2 \mid 1.0), (x_3 \mid 0.0), (x_4 \mid 0.6)\}$,
2. If $E = \{x_1, x_2, x_3, x_4\}$ and $M = [0, 1]$, find $\underset{\sim}{A}'$, $\underset{\sim}{B}'$ in 1(*i*) and (*ii*)
3. Prove DeMorgan's Laws for Fuzzy subsets.
4. Find $\underset{\sim}{A} \cdot \underset{\sim}{B}$ and $\underset{\sim}{A} + \underset{\sim}{B}$ in 1(*i*) and (*ii*).
5. Find the Fuzzy power set of the following :
   i) $S = \{x_1, x_2\}$ $M = \{0, 0.6, 1\}$
   ii) $S = \{x_1, x_2, x_3\}$, $M = \{0.5, 0.6\}$.

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Q If $A = \left\{\frac{0.3}{30}, \frac{0.7}{60}, \frac{1}{100}, \frac{0.2}{120}\right\}$ &
$B = \left\{\frac{0.2}{20}, \frac{0.4}{40}, \frac{0.6}{60}, \frac{0.8}{80}\right\}$ are fuzzy sets, then find the relation $R = A \times B$. Also find R^2.

Sol: $A = \left\{\frac{0.3}{30}, \frac{0.7}{60}, \frac{1}{100}, \frac{0.2}{120}\right\}$

$B = \left\{\frac{0.2}{20}, \frac{0.4}{40}, \frac{0.6}{60}, \frac{0.8}{80}\right\}$

Now $R_{A \times B} =$

	20	40	60	80
30	0.2	0.3	0.3	0.3
60	0.2	0.4	0.6	0.7
100	0.2	0.4	0.6	0.8
120	0.2	0.2	0.2	0.2

$R^2 = R \circ R =$

	20	40	60	80
30	0.2	0.3	0.3	0.3
60	0.2	0.4	0.6	0.7
100	0.2	0.4	0.6	0.8
120	0.2	0.2	0.2	0.2

$$\begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix}$$

$$R^2 = \begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix}$$

—x—

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(5'23)

Q If R and S are fuzzy relations given by

$$R = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{array}, \quad S = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.8 \\ 0.7 & 0.9 \end{bmatrix} \end{array}$$

Then find $R \cup S$, $R \cap S$, $\bar{R}$, $\bar{S}$.

Soln:- We know that

(1) $\mu_{R \cup S}(x,y) = \max(\mu_R(x,y), \mu_S(x,y))$

(2) $\mu_{R \cap S}(x,y) = \min(\mu_R(x,y), \mu_S(x,y))$

(3) $\mu_{R^c}(x,y) = 1 - \mu_R(x,y)$

$$R \cup S = \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.9 \\ 0.8 & 0.9 \end{bmatrix}$$

$$R \cap S = \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.8 \\ 0.7 & 0.6 \end{bmatrix}$$

$$\bar{R} = \begin{bmatrix} 0.5 & 0.9 \\ 0.8 & 0.1 \\ 0.2 & 0.4 \end{bmatrix}$$

$$\bar{S} = \begin{bmatrix} 0.4 & 0.5 \\ 0.6 & 0.2 \\ 0.3 & 0.1 \end{bmatrix}$$

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(Sh23)

Q Consider the fuzzy sets A, B, C defined on the interval $X = [0, 10]$ of integers by the membership grade function $\mu_A(x) = \frac{x}{x+2}$, $\mu_B(x) = 2^{-x}$, $\mu_C(x) = \frac{1}{1+10(x-2)^2}$. Determine the mathematical formulas for (i) $\bar{A}$ (ii) $A \cup B \cup C$.

Soln Given $X = [0, 10]$, $\mu_A(x) = \frac{x}{x+2}$, $\mu_B = 2^{-x}$, $\mu_C(x) = \frac{1}{1+10(x-2)^2}$

x	$\mu_A = \frac{x}{x+2}$	$\mu_B(x) = 2^{-x}$	$\mu_C(x) = \frac{1}{1+10(x-2)^2}$
0	0	1	$1/41 = 0.0244$
1	0.3333	0.5	0.0910
2	0.5	0.25	1
3	0.6	0.125	0.0910
4	0.6667	0.0625	0.0244
5	0.7143	0.0313	0.0110
6	0.75	0.0156	0.0006
7	0.7778	0.0008	0.0004
8	0.8	0.0004	0.0003
9	0.8182	0.0002	0.0002
10	0.8333	0.0010	0.0002

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$$A = \left\{ \frac{0}{0}, \frac{0.3333}{1}, \frac{0.5}{2}, \frac{0.6}{3}, \frac{0.6667}{4}, \frac{0.7143}{5}, \frac{0.75}{6}, \frac{0.7778}{7}, \frac{0.8}{8}, \frac{0.8182}{9}, \frac{0.8333}{10} \right\}$$

$$\overline{A} = \left\{ \frac{1}{0}, \frac{0.6667}{1}, \frac{0.5}{2}, \frac{0.4}{3}, \frac{0.3333}{4}, \frac{0.2857}{5}, \frac{0.25}{6}, \frac{0.2222}{7}, \frac{0.2}{8}, \frac{0.1818}{9}, \frac{0.1667}{10} \right\}$$

(iii) $A \cup B \cup C = \left\{ \frac{1}{0}, \frac{0.5}{1}, \frac{1}{2}, \frac{0.6}{3}, \frac{0.6667}{4}, \frac{0.7143}{5}, \frac{0.75}{6}, \frac{0.7778}{7}, \frac{0.8}{8}, \frac{0.8182}{9}, \frac{0.8333}{10} \right\}$

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